ISOTROPIC COORDINATES FOR GENERALIZED SCHWARZSCHILD-LIKE BLACK HOLES WITHIN ANISOTROPIC MEDIA

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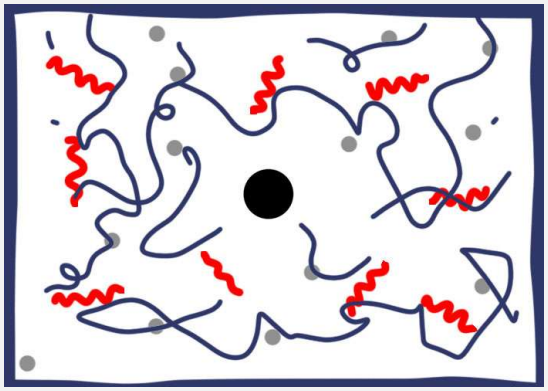
Plan

- Motivation
- Kiselev models as generalized Schwarzschild-like setup
- Main theorem: curvature → isotropic map
- Checks: Schwarzschild, Reissner-Nordstrom, Kottler
- Numerical relativity implications: two practical workflows
- Summary

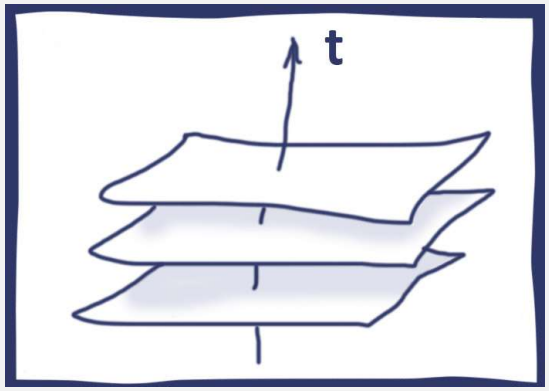
Motivation

Real astrophysical black holes are not in vacuum; environment affects ringdown, phasing, shadows, lensing.



Kiselev-type models are widely used as phenomenological “black holes + aniso-tropic fluid” backgrounds*.

For interpretation and for numerical relativity, a coordinate choice that is horizon-regular in the spatial sector is valuable.



* See **Domenica Garzon** presentation on **Mon, Mar 16, 2026, 04:42 PM**
APR-D94:2: Scattering of Scalar Fields in Charged Black Holes Surrounded by an Anisotropic Fluid

A black hole in the mixture of anisotropic fluids

Anisotropic perfect fluids, static spacetime (Kiselev class)

$$ds^2 = f(r)dt^2 - f(r)^{-1}dr^2 - r^2 d\sigma^2$$

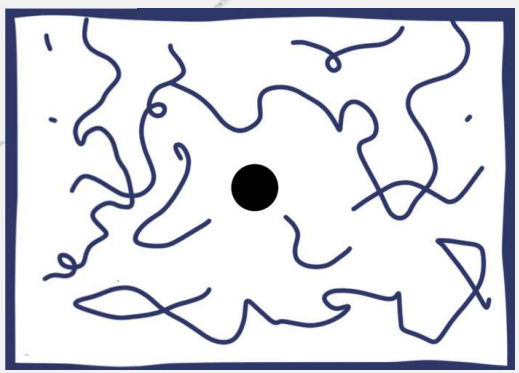
$$f(r) = 1 - \frac{r_g}{r} - \sum_{\mathrm{i}} \frac{K_{\mathrm{i}}}{r^{3w_{\mathrm{i}}+1}}$$

$$\frac{p_{r(\mathrm{i})} + 2p_{\perp(\mathrm{i})}}{3} = w_{\mathrm{i}}\rho_{\mathrm{i}}$$ Effective equation of state

w	Source
1/3	Radiation
-1	Vacuum
-1/3	Strings
-2/3	Domain Walls

Particular cases:
Schwarzschild solution $K = 0$
Reissner-Nordstrom solution $K=-Q^2$
Kottler (Schwarzschild-de Sitter solution) $K=\Lambda/3$
Reissner-Nordstrom-de Sitter solution $K_1=-Q^2$, $K_2=\Lambda/3$

$$M, Q, K, w$$



Superradiance in Cosmology
(ongoing project)

Curvature to Isotropic coordinates map – multi-index theorem

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\sigma^2 \rightarrow ds^2 = -F(\rho)dt^2 + \Phi(\rho)^4(d\rho^2 + \rho^2 d\sigma^2)$$

$$\boxed{\rho(r) = \rho_0\, r\, e^{I(r)} \qquad I(r) = \int_{r_+}^{r} \frac{dx}{x}\left(\frac{1}{\sqrt{f(x)}} - 1\right)}$$

$$f(r) = 1 - \xi(r) = 1 - \sum_{\mathrm{i}} \xi_{\mathrm{i}}(r)\,, \qquad \xi_0(r) = r_g/r\,, \qquad \xi_{\mathrm{i}}(r) = K_{\mathrm{i}}/r^{s_{\mathrm{i}}}$$

$$\frac{1}{\sqrt{f(x)}} = \frac{1}{\sqrt{1-\xi(x)}} = \sum_{k=0}^{\infty} \frac{(2k)!}{4^k (k!)^2}\, \xi(x)^k$$

place to watch for truncation I

$$\xi^k = (\xi_0 + \cdots + \xi_N)^k = \sum_{\mathbf{m}} \frac{|\mathbf{m}|!}{\prod_{\mathrm{i}=0}^{N}(m_{\mathrm{i}}!)} \prod_{\mathrm{i}=0}^{N} \xi_{\mathrm{i}}^{m_{\mathrm{i}}}$$

Multinomial theorem

index list $\mathbf{m} = (m_0, m_1, \ldots, m_{\mathrm{N}})$,

$|\mathbf{m}| = m_0 + m_1 + \cdots + m_{\mathrm{N}} = k$

$$\boxed{I(r) = \int_{r_+}^{r} \frac{dx}{x}\left(\frac{1}{\sqrt{f(x)}} - 1\right) = \sum_{|\mathbf{m}|=k,\, k=1} C(\mathbf{m})\, r_g^{m_0} \prod_{\mathrm{i}=1}^{n} K_{\mathrm{i}}^{m_{\mathrm{i}}} \int_{r_+}^{r} dx\, x^{-1-\alpha(\mathbf{m})}}$$

$$C(\mathbf{m}) = \frac{(2|\mathbf{m}|)!}{4^{|\mathbf{m}|}(|\mathbf{m}|!)^2} \frac{|\mathbf{m}|!}{\prod_{\mathrm{i}=0}^{n}(m_{\mathrm{i}}!)} \qquad \alpha(\mathbf{m}) = m_0 + \sum_{\mathrm{i}=1}^{n} m_{\mathrm{i}} s_{\mathrm{i}}$$

Classic cases recovered

Schwarzschild solution $$r(\rho) = \rho\left(1 + \frac{r_g}{4\rho}\right)^2$$

Reissner-Nordstrom solution $$r(\rho) = \rho\left(1 + \frac{r_g}{2\rho} + \frac{r_g^2 - 4Q^2}{16\rho^2}\right)$$

Kottler (Schwarzschild-De Sitter) solution - isotropic map matches the elliptic-integral/Jacobi form known in the literature R. Solanki, Class. Quant. Grav. 39, 015015 (2022), arXiv:2103.10002 [gr-qc]

Lagrange–Bürmann local inversion

$$\rho_\star \coloneqq \rho(r_\star) \qquad \Psi(r) \coloneqq \rho(r)$$

$$r(\rho) = r_\star + \sum_{n=1}^{\infty} \frac{(\rho-\rho_\star)^n}{n!} \left[\frac{\mathrm{d}^{\,n-1}}{\mathrm{d}r^{\,n-1}} \left(\frac{r-r_\star}{\Psi(r)-\Psi(r_\star)} \right)^n \right]_{r=r_\star} = r_\star + \sum_{n=1}^{\infty} c_n (\rho-\rho_\star)^n$$

place to watch
for truncation II

Converges in the open disk: $|\rho-\rho_\star| < R, \qquad R = \mathrm{dist}\big(\rho_\star, \Sigma_\rho\big)$

Workflow for Numerical Relativity

Route A: Integration + root-finder (global, high accuracy)

- Compute the forward map $\boldsymbol{\rho}(\boldsymbol{r}) = \boldsymbol{\rho}_0\, \boldsymbol{r}\, \mathrm{e}^{\boldsymbol{I}(\boldsymbol{r})}$ (numerically, or with a truncated multi-index series for $\boldsymbol{I}(\boldsymbol{r})$).
- For a given $\boldsymbol{\rho}$, solve $\boldsymbol{\rho}(\boldsymbol{r}) - \boldsymbol{\rho} = 0$ via Newton-Raphson to obtain $\boldsymbol{r}(\boldsymbol{\rho})$.
- Best when you need **very high accuracy** or only a **few scattered** evaluations.

Route B: Analytic inverse + double truncation (fast on grids)

- Build a local inverse series $\boldsymbol{r}(\boldsymbol{\rho})$ around a base point $\boldsymbol{\rho}_\star = \boldsymbol{\rho}(\boldsymbol{r}_\star)$ using Lagrange-Burmann inversion.
- Truncate $\boldsymbol{I}(\boldsymbol{r})$ at $|\boldsymbol{m}| \leq \boldsymbol{k}_{max}$, $\boldsymbol{r}(\boldsymbol{\rho})$ at $\boldsymbol{N}$ terms
- Best when you need **many evaluations on a regular grid** and want analytic derivatives for low cost

In practice, a hybrid strategy works best:

- use the Lagrange series for fast, analytic grid evaluation,
- then refine with Newton–Raphson where maximum accuracy is required, e.g., in constraint violation checks or horizon location.

Error control

When using Route B, total error can be decomposed schematically

$$|\boldsymbol{r}(\boldsymbol{\rho}) - \boldsymbol{r}_{\mathrm{true}}(\boldsymbol{\rho})| \leq \boldsymbol{\varepsilon}_{\boldsymbol{I}}^{(\boldsymbol{k}_{max})} + \boldsymbol{\varepsilon}_{\boldsymbol{L}}^{(\boldsymbol{N})} + \boldsymbol{\varepsilon}_{\mathrm{coupling}}$$

Decay geometrically.

Balancing errors gives a practical rule: $\boldsymbol{N} \approx \mathbf{1.5} \times \boldsymbol{k}_{\mathbf{max}}$.

Examples:

- Schwarzschild at $\boldsymbol{k}_{\mathbf{max}} = \mathbf{20}$, $\boldsymbol{N} = \mathbf{32}$: combined error $\sim\mathbf{10}^{-9}$
- Reissner–Nordström ($\boldsymbol{Q}^2 = \mathbf{0.5}\ \boldsymbol{r}_g^2$) at $\boldsymbol{k}_{\mathbf{max}} = \mathbf{25}$, $\boldsymbol{N} = \mathbf{30}$: error $\sim\mathbf{10}^{-9}$
- Kottler at $\boldsymbol{k}_{\mathbf{max}} = \mathbf{30}$, $\boldsymbol{N} = \mathbf{20}$: error $\sim\mathbf{10}^{-6}$ (sufficient for initial-data construction)

Summary

- **General map**: A constructive isotropic-coordinate transformation for a broad Schwarzschild-like / Kiselev-type class.
- **Validated**: recovers Schwarzschild, Reissner-Nordstrom, Kottler.
- **Practical use**: directly enables conformally flat initial data and clean environmental parameter studies; two computational routes (root-finder vs analytic inverse) cover *high-accuracy sparse* and *fast grid* regimes.

THANK YOU